

Practical Practice Program List

1. Write a program to calculate simple interest.
2. C Program to Multiply two Floating-Point Number
3. C Program to Find the Size of int, float, double, and char using sizeof().
4. Any year entered through the keyboard. Write a program to determine whether the year is leap year or not. Use the logical operator && and ||.
5. Write a C program to enter any character and check whether character is alphabet, digit or any special character using 'if...else'.
6. Write a program to find the entered character is uppercase or lowercase.
7. Write a program to check whether a triangle is valid or not, when the lengths of three sides of the triangle are entered through the keyboard. Also check the triangle is equilateral, isosceles, scalene or right-angled triangle.
8. C program to print number of days in a month using else if ladder.
9. Write a C program to accept a coordinate point in a XY coordinate system and determine in which quadrant the coordinate point lies.
10. Write a C program to input basic salary of an employee and calculate its Gross salary according to following rule :

Basic salary $\leq$ 10000: HRA = 20%, DA = 80%

Basic salary range (10001–20000) : HRA = 25%, DA = 90%

Basic Salary range(20001–30000) : HRA = 30%, DA = 95%

[gross salary=Basic salary + HRA + DA]

11. A certain grade of steel is graded according to the following conditions :

- (1) Hardness must be greater than 50.
- (2) Carbon contents must be less than 0.7.
- (3) Tensile strength must be greater than 500.

The grades are as follows :

Grade is 10 if all three conditions are met.

Grade is 9 if conditions (i) and (ii) are met.

Grade is 8 if condition (ii) and (iii) are met.

Grade is 7 if condition (i) and (iii) are met.

Grade is 6 if only one condition is met.

Grade is 5 if none of the conditions are met.

12. A library charges a fine for every book returned late.

For first 5 days the fine is 50 paise,

for 6-10 days fine is one rupee and

above 10 days fine is 5 rupees.

If you return book after 30 days your membership will be cancelled.

Write a program to accept a number of days the member is late to return the book and display the fine or the appropriate message.

13. A cloth showroom has announced the following festival discounts on the purchase of items based on the total cost of the item purchased :

Total cost discount

Up to 2000 5%

Rs. 2001 to rs. 5000 25%

Rs. 5001 to rs. 10000 35%

Above Rs. 10000 50%

Write a program to input the total cost of purchase and display the final paid amount.

14. Write a program to calculate Employee Loan using if-else statement.

Category Name	Marital Status	Service	Loan
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(1) Permanent	Unmarried	≥ 30	50000
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Else 25000

(2) Permanent	Married	≥ 30	40000
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Else 20000

(3) Temporary	Married/Unmarried/ N. A.		10000
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15. Admission to a college in science branch is given if the following conditions Area satisfied

(i) Maths marks ≥ 80

(ii) Physics marks ≥ 75

(iii) Chemistry marks ≥ 70

(iv) Total percentage in all three subjects ≥ 80

Given the marks in three subjects, write a program to process the applications to list the eligible candidates.

16. A telephone bill is charged as per following rules :

- first 150 calls are free of charge.
- Next 200 calls are charged @ Rs. 0.80 per call.
- Next 500 calls are charged @ Rs. 1.00 per call
- Next 200 calls are charged @ Rs. 1.20 per call
- Each next call is charged @ Rs. 1.50 per call.

Telephone line rent is Rs. 250/- irrespective of numbers of calls dialed.

Write a program to accept telephone calls and calculate and display total bill payable.

17. Write a menu driven program for following cases :

- (1) To calculate area of a Circle.
- (2) To calculate area of a Rectangle.
- (3) To calculate area of a Triangle.
- (4) Exit.

18. Write a menu driven program which has following options:—

- (1) To swap two numbers using third variables.
- (2) To calculate area of Triangle.
- (3) To check entered number is positive or negative using bitwise operator.
- (4) Exit.

Once a menu item is selected the appropriate action should be taken and once this action is finished, the menu should reappear. Unless the user selects the 'Exit' option the program should continue to work.

19. Using switch–case control structure write a program to print a given integer number as a string. e–g. integer 2 3 4 9 should be printed as "TWO THREE FOUR NINE".

20. Write a program which take lowercase alphabet as input and determine whether it is vowel or consonant or last alphabet using switch statement

21. Write menu driven program using switch case statement to generate arithmetic calculator i. e. (+ , – , * , /) operations on two inputted number.

22. Write a menu driven program which has following options :

- (1) Factorial of a program.
- (2) Prime or not.
- (3) Odd or even.
- (4) Exit.

Once a menu item is selected the appropriate action should be taken and once this action is finished, the menu should reappear.

23. Write a program to convert decimal number into binary number.

24. Write a c program to check whether a number is strong or not.

(A number is called strong number if sum of the factorial of its digit is equal to number itself.
For example : 145 since $1! + 4! + 5! = 1 + 24 + 120 = 145$)

25. If value of 'n' is input through keyboard, then write a program to compute and display 'n' number of prime numbers from 'n' onwards.

26. Identify the different test cases for the prime number program which print prime number between 1 to 100 intervals and validate the design test cases ?

27. Write a C program to reverse a number using while loop

28. Write a C program to print numbers from 1 to N.

29. Print Square, Cube and Square Root using C program

30. Write a program to find factors of a number.

31. Write a program to print out all Armstrong numbers between 1 and 500. If sum of cubes of each digit of the number is equal to the number itself, then the number is called an Armstrong number. {For example,

$$153 = (1 * 1 * 1) + (5 * 5 * 5) + (3 * 3 * 3)}$$

32. Write a program to find Perfect number.

(6 is a perfect number, Sum of its proper divisors = $1 + 2 + 3 = 6$.)

33. Write a C program to print the sum of even and odd numbers separately

34. Write a C program to print Fibonacci series upto 'n' terms.

35. Write a program to print the factorial of any entered number using loop.

36. Enter a number from the keyboard and then calculate the no of digits and sum of digits of that number using a while loop.

37. Write a loop that will calculate the sum of every third integer beginning with $i = 2$ (i-e. calculate the sum of $2+5+8+11+---$) for all values of i that are less than 100. Write the loop in two ways:

(i) Using a do---- while statement.

(ii) Using a for statement.

38. Print the numbers in the following format up to a given number and that number is entered from the keyboard.

2 4 6 8 up to that given number

39. Write syntax and flowchart of do.. while loop control statements and Write a C program to compute the sum of first n terms ($n \geq 1$) of the following series using 'for' loop.

$$1 - 3 + 5 - 7 + 9 - \dots$$

40. Write a program to find n th term of the sequence 1,10,101,1000, 10001, where n is input by user.

41. Write a program to add "n" terms of the following series using loop :

$$\text{Sum} = 1 + 1/x + 1/x^2 + 1/x^3 + \dots + N$$

42. Write a program in C to display the cube of the number upto given an integer.

43. Write syntax and flowchart of while loop control statements And Write a program to print following pyramid.

12345

12345

12345

12345

44. Write a C program to display following pattern

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45. Write a C program to display Floyd's triangle

1

2 3

4 5 6

7 8 9 10

46. Why Indentation and Comments required in 'C' programs ? Write a C program to display following pattern.

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47. Write a program to print following pyramid :

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48. Write a C program to display following pattern.

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

49. Write a program in C to make such a pattern like right angle triangle with a number which will repeat a number in a row.

The pattern like :

```
1
1 2
1 2 3
1 2 3 4
```

50. Write a program to print following pyramid :

```
A B C D E F E D C B A
A B C D E D C B A
A B C D C B A
A B C B A
A B A
A
```

51. Write a program to print following pyramid :

```
A
A B A
A B C B A
A B C D C B A
```

52. Write a program in C to display the sum of the series [$1+x+x^2/2!+x^3/3!+....$].

53. Write a program to print following pyramid :

```
* * * * *
* * * *
* * *
*
```

54. Write a program in C to display the sum of the series [9 + 99 + 999 + 9999 ...].

55. Write a program to print following pyramid :

```
1
2 3 2
3 4 5 4 3
4 5 6 7 6 5 4
5 6 7 8 9 8 7 6 5
```

56. Write a function to exchange values of two variables.

57. Write the user defines function to reverse a given number and check whether it is palindrome or not.

58. Write a function topper () to convert lower case letter to upper case.

59. Write a program to calculate average of 'n' numbers by using user defined function.

60. Write a function to implement binary search

61. Write a function to implement a simple calculator (Add, Subtract, Multiply, Divide)

62. Program to print perfect numbers in given range using functions

(28 is a perfect number, Sum of its proper divisors = 1 + 2 + 4+7+14 = 28.)\

63. Write a function which will return 1 if number supplied to the function is prime and return 0, if number supplied to the function is non prime. Using this function display all prime numbers in the range 1 to 100.

64. Write a program to check whether the entered number is single digit or more digit, by using user defined function.

65. Write a program that returns the sum of squares of all odd numbers from 1 to 50, by using function.

66. Write a program to find out the HCF of two numbers by using user defined function.

67. Write a program using function which accept the age of Ram, Sham, Krishna as integer input and display youngest amongst them.

68. Write a function prodDigits () that inputs a number and returns the product of digits of that number.

69. Write a C program to find the factors of a given integer by using function.

70. Write a function to find greatest of three numbers.

71. Write a recursive function to find factorial of a number.

72. Write the user-defined function for converting decimal number to binary

73. Write a program by using function that prints a given positive integer in reverse order and also generate the sum of individual digits of resultant reversed integer.

- 74.** Develop a recursive function to find the sum of digits of a given integer.
- 75.** Write a recursive function to display starting n terms of Fibonacci series.
- 76.** Write a C program to find Sum of Digits of a Number using Recursion.
- 77.** Write a C program to reverse a Number Using Recursion
- 78.** Write a C program to do the Sum of First N Natural Numbers Using Recursion
- 79.** Find GCD (Greatest Common Divisor) Using Recursion.
- 80.** Write a program to perform binary search through recursion.
- 81.** Write a program to find 'a' raised to power 'n' (a^n) by using recursion where the value of 'a' & 'n' are entered by the user.
- 82.** Write a C program to enter elements for 1 - D array and find sum of all elements.
- 83.** Write a program to find the sum and average of 10 real numbers in an array.
- 84.** Write a C program to sort the given array elements in descending order using Bubble sort.
- 85.** Write a C program to sort the given array elements in ascending order using Bubble sort.
- 86.** Write a C Program to input elements for 1 – D array and Calculate Sum and Average of an Array elements.
- 87.** Write a C program to enter elements for 1–D Array and insert an element in a specified position in a given array.
- 88.** Write a C program to enter elements for 1–D Array and find Largest and Smallest Element in Array
- 89.** Write a C program to find Second Largest Element in an Array.
- 90.** Write a C program to enter elements for 1-D array and find sum of all elements.
- 91.** Write a program to input 'n' elements for 1–D array and search an element using Linear search.
- 92.** Write a program to input 'n' elements for 1–D array and search an element using Binary search.
- 93.** Write a C Program to Check Whether Two Matrices Are Equal or Not.
- 94.** Write a C program to input a 3×3 matrix and print the sum of rows.
- 95.** If two 4*4 matrices are given input from console then write a program to perform and display multiplication of these two matrices.
- 96.** Write a C program to multiply two matrices of different order.
- 97.** Write a C program to enter elements for 2-D Matrix and print the sum of diagonal elements of a matrix.
- 98.** Write a C program to enter elements for 2-D Matrix. Convert that matrix into upper triangular and lower triangular matrix.

99. Write a program in C to accept a matrix and determine whether it is a sparse matrix.

/*A sparse matrix is matrix which has more zero elements than nonzero elements */

100. Write a program in C to accept a matrix and determine whether it Magic matrix.

(A Magic Square is a $n \times n$ matrix of the distinct elements from 1 to n^2 where the sum of any row, column, or diagonal is always equal to the same number.)